

Unit 10, Lesson 5: Matching Data Displays

Benchmark

MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts.

Learning Target

- ☐ I can match displays of data to the appropriate data set.

10.5.1 Warm-Up: Bell Work

1. During the 2020 NBA Season, Jimmy Butler played in the championship game against the Los Angeles Lakers. The total points scored by Jimmy Butler for the conference semifinals, conference finals, and championship games are listed below.

12, 35, 22, 40, 25, 23, 22, 17, 24, 17, 14, 20, 17, 17, 30, 13, 40

- a. Determine the five-number summary for the data set.

- b. Determine the range.

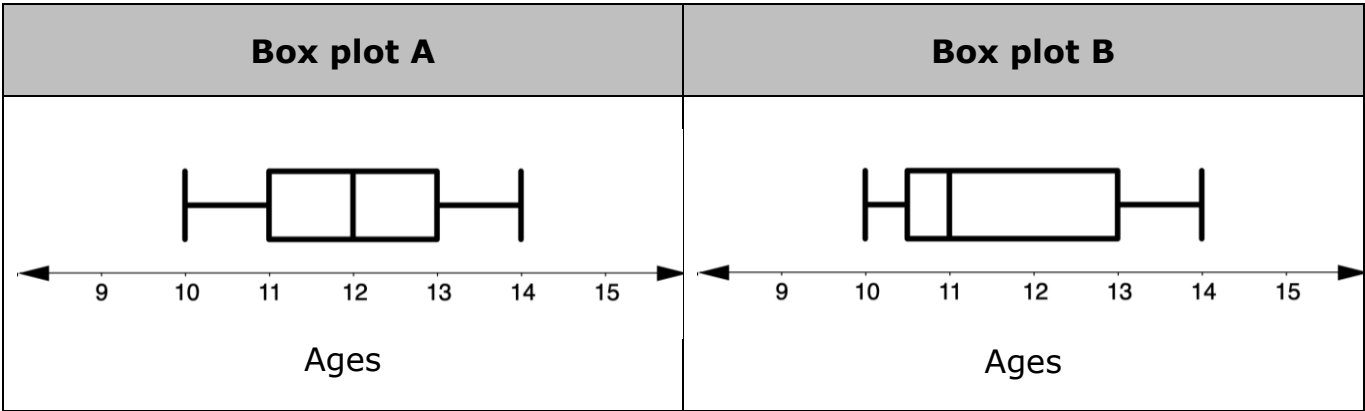
10.5.2 Exploration: Matching Displays

Work with your partner to complete the following.

A middle school band director collects the ages of each clarinet player in the band, shown below.

10, 10, 11, 11, 11, 12, 12, 12, 13, 13, 14

1. Determine the five-number summary and IQR for the ages of the clarinet players in the band.
2. Which box plot accurately displays the ages of the clarinet players?



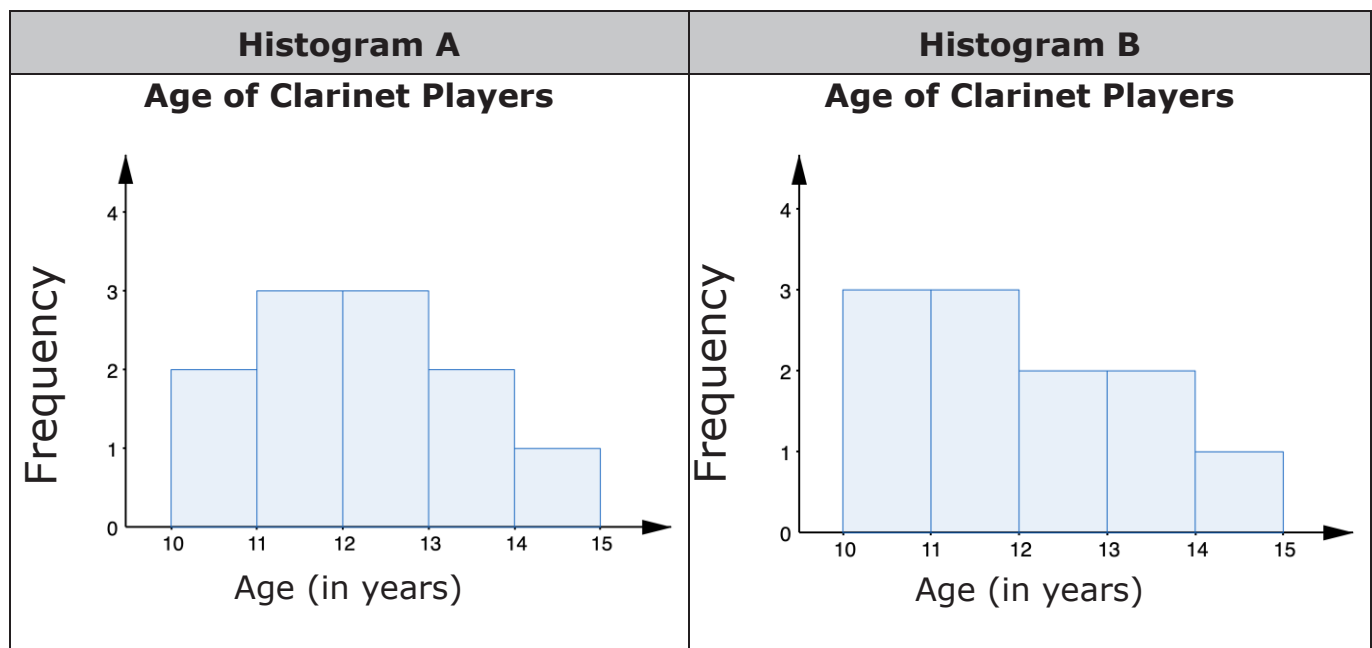
3. Discuss your choice with your partner and summarize.

In a box plot, whiskers extend from each quartile to the minimum or maximum. The left whisker represents the bottom 25% of the data and the right whisker represents the top 25%.

4. Discuss with your partner how the location of the median and the length of the whiskers can be used to determine the shape of the distribution. Summarize your discussion.
5. Create a frequency table of the data set.

Age Range	Number of Clarinet Players

6. Which histogram most accurately displays the data?

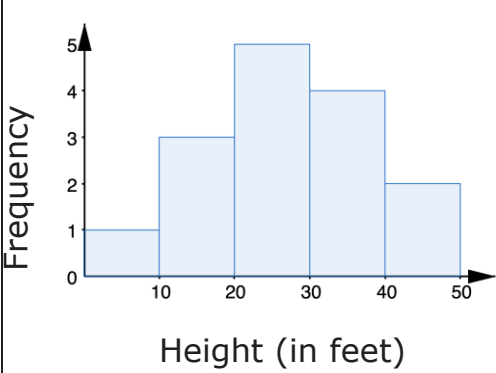
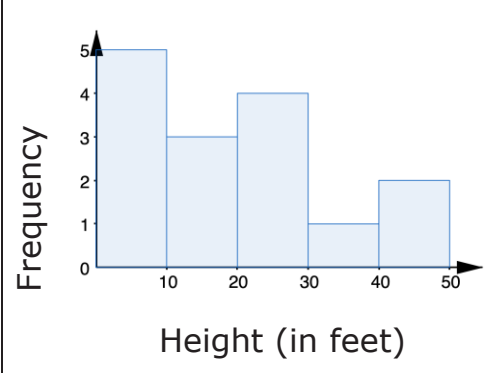
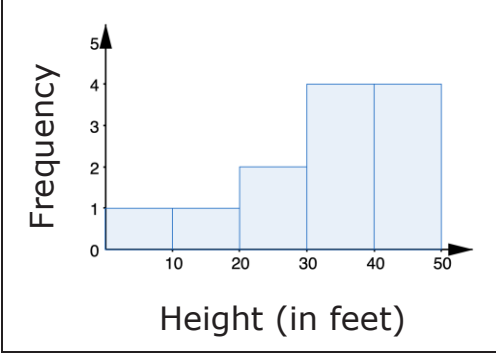


7. Discuss which characteristics of the histogram you used to make your selection.
8. Describe the shape of the histogram you chose, including any gaps, outliers, or clusters.
9. Compare the box plot selected for the data with the histogram selected. Discuss which display better represents the data. Summarize your discussion.

10.5.3 Activity: Card Sort

1. An arborist visited a pine tree farm and took measurements of the tree heights, in feet, in three different fields. He dropped his paperwork and all of the data he collected got mixed up.

Complete the table to match each histogram with its corresponding data set and box plot. Write the number and letter from the card that corresponds to each match.

Histogram	Data Set	Box Plot
		
		
		

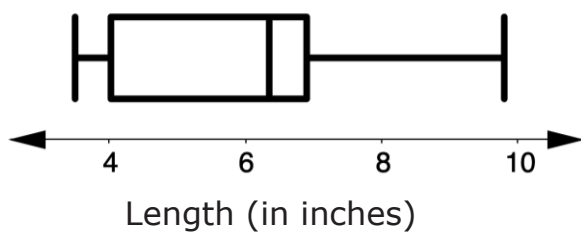
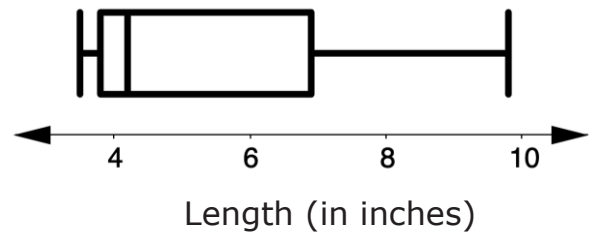
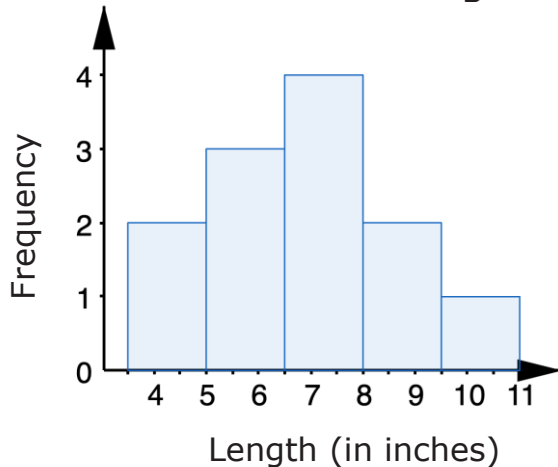
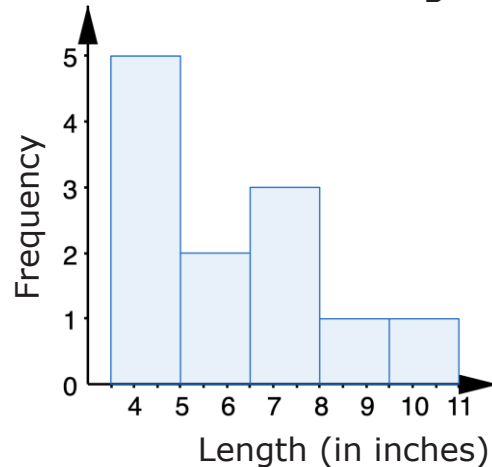
2. Compare your answers with your partner. Make any revisions, if needed.

10.5.4 Your Turn!

1. A horticulturist is researching sunflowers to find out the lengths of the petals on a sunflower. She gathers information from 12 sunflowers and measures the length of the petals, in inches, on each flower.

6.8, 6.4, 6.2, 3.5, 3.6, 4.1, 3.9, 4.5, 6.7, 7.2, 8.1, 9.8

Circle the boxplot and the histogram that best display the data set.

Sunflower Petal Lengths**Sunflower Petal Lengths****Sunflower Petal Lengths****Sunflower Petal Lengths**

10.5.5 Wrap-Up: Growth Mindset

1. Complete the table as you reflect on today's lesson.

Which mindset did I demonstrate?	Choose the best description.
Did I use the partner discussions as learning opportunities?	○ Never ○ Sometimes ○ Usually ○ Always
Did I work on problems that challenged me?	○ Never ○ Sometimes ○ Usually ○ Always
Did I use strategies to “un-stick” myself when I found the problems difficult?	○ Never ○ Sometimes ○ Usually ○ Always